RESEARCH PAPER ANALYSIS

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins

Source: arxiv.org/abs/2504.04715

THE CORE PROBLEM

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

The Threat

Commercial LLM API providers charge premium rates for specific models (e.g., Llama-3-70B, Qwen2-72B), but the black-box nature of APIs means users have **no guarantee** the advertised model is actually being served.

Providers face **strong economic incentives** to covertly substitute cheaper alternatives—quantized versions, smaller models, or entirely different architectures—to reduce GPU costs while maintaining pricing.

Formal Framework

The problem is formalized as hypothesis testing:

$$H_0: P\_spec(y|x)$$
vs
$$H_1: P\_actual(y|x)$$

Given samples from the API, can an auditor distinguish the specified model's distribution from an alternative?

EMPIRICAL EVIDENCE

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Five model families evaluated comparing original precision against FP8 quantization across four benchmarks. Small performance differences make detection extremely challenging.

MODEL	MMLU	GSM8K	MATH	GPQA DIAMOND
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct FP8	62.43 ±0.26	60.90 ±4.10	14.91 -5.74	20.14 -2.48
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct FP8	77.88 ±0.13	87.35 ±1.24	35.75 ±1.16	33.12 ±0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it FP8	71.92 ±0.11	79.41 ±1.14	32.53 ±0.34	27.81 ±3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct FP8	81.98 ±0.08	86.82 ±0.97	37.67 ±1.39	31.08 ±1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3 FP8	58.77 ±0.13	32.20 ±4.02	7.68 ±1.23	22.72 ±0.19

DETECTION CHALLENGE

Higher Temperature Amplifies Performance Gaps but Increases Noise

Temperature vs Accuracy Analysis

Temperature scaling experiments (temperatures 0.2, 0.5, 1.0, 2.0) show that higher temperatures degrade all models' performance.

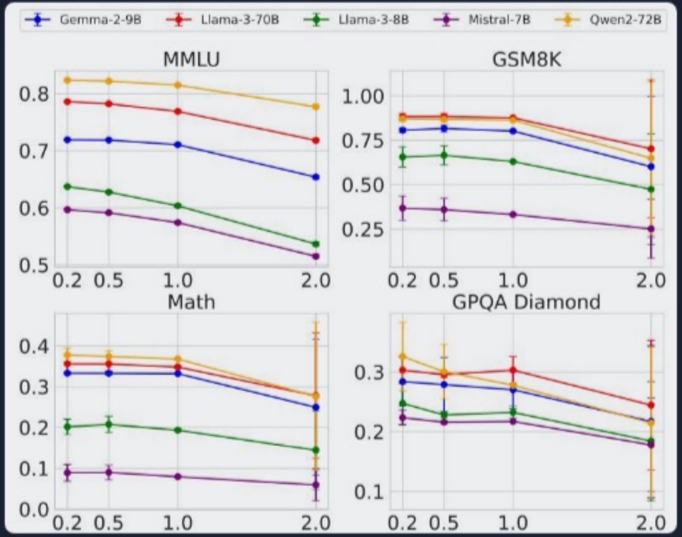

Figure 1: Benchmark accuracy vs temperature for five models across four tasks

The Detection Problem

At temperature 2.0:

- Performance degrades across all models
- The gap between original and substitute **narrows**
- Variance increases significantly
- Statistical signal becomes **noisy and unreliable**

The substitution signal that might distinguish models at low temperature becomes obscured at production-relevant temperatures.

DETECTION FAILURE

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

The Approach

A promising detection approach compares token-level log probabilities between a local reference model and the API response.

The assumption: different models produce different log probability distributions for the same inputs.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure 5: Log probability variation across inference frameworks and hardware

ADVERSARIAL ATTACK

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

The Attack Vector

Providers can implement **randomized substitution**: routing only a fraction *p* of queries to the cheaper substitute model.

User Query

↓

Random Routing
Probability p → substitute
Probability (1-p) → genuine

↓

API Response

Mathematical Reality

As *p* approaches 0, the mixed distribution converges to the specified model's distribution:

$$P_{mixed} = (1-p) \times P_{spec} + p \times P_{sub}$$

$$\text{as } p \to 0\text{: } P_{mixed} \approx P_{spec}$$

→ 0

DETECTION POWER

∞

QUERIES NEEDED

This makes software-based detection **fundamentally query-prohibitive** against realistic adversarial providers.

Additional Countermeasure: Benchmark evasion — providers can detect known benchmark prompts and route them to the genuine model while serving substitutes for regular traffic.

DETECTION FAILURES

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Approach	Critical Weakness	Practical Limitation
Text-Based Statistical Tests	MMD (Maximum Mean Discrepancy) and similar tests on output distributions	QUERY-INTENSIVE Requires impractically large query budgets; fails against subtle or randomized substitutions	Cannot distinguish quantization noise from actual model differences; statistical power collapses at low substitution rates
Log-Probability Fingerprinting	Compare token-level log probabilities between local reference and API responses	NONDETERMINISM Defeated by production-level variation across software versions and GPU hardware	Same model on different infrastructure produces different log probabilities; false positives/negatives are unavoidable
Adversarial Countermeasures	Provider-side evasion: randomized routing, benchmark detection	ACTIVE DEFENSE Makes all passive detection fundamentally query-prohibitive	Mixed distribution converges to target as substitution probability decreases; auditor cannot obtain enough statistical power

THE SOLUTION

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

How TEEs Work

Trusted Execution Environments (TEEs) are hardware-secured enclaves that provide **cryptographic attestation** of the code and data being executed.

- Prove that specific model weights are loaded
- Attest that correct inference code is running
- Generate verifiable evidence of execution integrity
- Only modest performance overhead

Unlike ZKPs which are computationally prohibitive for large models, TEEs are **practical and deployable today.**

TEE Properties vs Software Methods

PROPERTY	SOFTWARE METHODS	TEES
Provable	NO	YES
Efficient	YES	YES
Robust to Evasion	NO	YES
Production-Ready	PARTIAL	YES

SUMMARY

Key Takeaways: Close the Verification Gap with Hardware Attestation

1. **Model substitution is economically motivated and practically undetectable via software.** Providers have clear financial incentives to serve cheaper models, and statistical detection methods cannot keep pace with adversarial countermeasures.

2. **FP8 quantization barely affects benchmarks.** Most performance drops fall within error bars, making it an attractive and nearly undetectable cost-saving substitution for providers.

3. **Inference nondeterminism kills log-probability fingerprinting.** Production environments introduce variation across software versions and GPU hardware that makes metadata-based detection unreliable.